

2(a)(i)	<i>idea of</i> compare / AW, observed and expected (data / results / ratios / phenotypes) or data is categoric / discrete / discontinuous / nominal ;	1
2(a)(ii)	<i>idea of</i> no (significant) difference between the observed and the expected (results) or <i>idea of</i> difference between observed and expected (results) is due to chance ;	1

Question	Answer				Marks
2(a)(iii)	offspring phenotype	O	E	$\frac{(O - E)^2}{E}$	3
	purple axial flowers	1756	1882 (A 1881) (1881.56250)	8.44 (8.379175)	
	purple, terminal flowers	653	627 (627.18750)	1.08 (1.062338)	
	white, axial flowers	702	627 (627.18750)	8.97 (8.923823)	
	white, terminal flowers	234	209 ; (209.06250)	2.99 ; (2.974608)	
	$\chi^2 =$			21.48 ; (21.339944)	
	$\chi^2 = 21.33 / 21.34 / 21.48$ for mp3 (to allow for using calculator values or rounded values)				
2(a)(iv)	<u>7.82</u> ;				1

Question	Answer	Marks
2(a)(v)	<i>apply ecf as needed</i> 1 reject (null hypothesis) <u>and</u> state that, the calculated value / χ^2, is higher than, 7.82 / the critical value / the critical value they give in (iv) <u>from the table</u> ; 2 <i>If null hypothesis rejected:</i> (between O and E): there is a significant difference / the difference is not due to chance <i>idea that</i> there is a less than 5% probability that the difference is due to chance / there is a more than 95% probability that the difference is not due to chance ; 3 <i>From mp1 If null hypothesis rejected or mp2 If there is a significant difference / AW stated</i> <i>idea that:</i> genes are on the same chromosome / are not on different chromosomes / are linked / show autosomal linkage / do not assort independently ; A <i>idea of, 'may be / can be', throughout</i>	3